DAMIEN CHEMISTRY OLYMPIAD NOTES

Atomic Structure

Question Bank without Worked Answers

Core set: 24 concepts × 3 levels = 72 questions
Extension set: 6 mass-spectrometry and 8 photoelectron-spectroscopy questions

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1. Atomic Composition

1.1 Proton, neutron and electron numbers in atoms and ions

1A Easy

An ion has the symbol $^{27}_{13}Al^{3+}$.
Determine the number of:
1. protons,
2. neutrons,
3. electrons.

1B Medium

An ion X^{2-} contains 18 electrons and 16 neutrons.

Determine:
1. the proton number of X,
2. the nucleon number of the ion,
3. the identity of X.

1C Hard

The ion X^{2+} contains 18 electrons and 22 neutrons. The ion Y^{-} has the same nucleon number as X^{2+} and contains 20 electrons. Determine the nuclide notation of both ions.

1.2 Proton number, nucleon number and isotopes

2A Easy

Define:
1. proton number,
2. nucleon number,
3. isotope.

2B Medium

Two atoms are represented by $^{63}_{29}X$ and $^{65}_{29}X$.
Explain:
1. why they are isotopes,
2. which subatomic particle accounts for their different masses,
3. why their chemical properties are very similar.

2C Hard

Two neutral species P and Q satisfy the following conditions:
- P and Q have the same number of neutrons.
- P has one more proton than Q.
- P has a nucleon number one greater than Q.

Are P and Q isotopes? Explain whether they are likely to have similar chemical properties.

1.3 Relative charge and relative mass

3A Easy

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State the relative charge and relative mass of a proton, neutron and electron.

3B Medium

Arrange the following particles in increasing magnitude of charge-to-mass ratio: $^{4}_{2}He^{2+}$, $^{1}_{1}H^{+}$, $^{2}_{1}H^{+}$, e^{-}. Explain your ordering.

3C Hard — Multi-statement MCQ

A proton, an electron and an alpha particle are placed separately in the same uniform electric field.
1. The alpha particle has a smaller charge-to-mass ratio than the proton.
2. The proton and electron experience accelerations of equal magnitude.
3. The particle with the largest charge-to-mass ratio experiences the largest acceleration magnitude.

Which statements are correct? A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1 and 3 only E. 1, 2 and 3

1.4 Behaviour of charged particles in electric fields

4A Easy

A positively charged ion enters a uniform electric field directed from left to right.
State:
1. the direction of the force on the ion,
2. the direction of the force on an electron in the same field.

4B Medium

A proton and an alpha particle are initially at rest in the same uniform electric field. Taking the mass of the alpha particle to be four times the mass of a proton, calculate $\frac{a_{\text{proton}}}{a_{\text{alpha}}}$.

4C Hard

An α particle is accelerated through a perpendicular electric field and is deflected with an angle of $+2°$. If this is now replaced with an Cl^{-} particle, what is the angle to which it is deflected by? Assume angle $\vartheta \propto \frac{q}{m}$.

1.5 Distribution of mass and charge within an atom

5A Easy

Describe how mass and charge are distributed within an atom.

5B Medium

In Rutherford's alpha-particle scattering experiment:
1. most alpha particles passed through the foil with little or no deflection,
2. a small fraction were strongly deflected,
3. a very small fraction travelled backwards.

Explain what each observation reveals about atomic structure.

5C Hard

Suppose the positive charge of an atom were spread uniformly throughout the entire volume of the atom rather than concentrated in a nucleus. Predict how Rutherford's scattering results would differ.

1.6 Relative atomic mass from isotopic abundance

6A Easy

An element consists of 80.0% isotope ^{24}X and 20.0% isotope ^{26}X. Calculate its relative atomic mass.

6B Medium

Chlorine consists only of 35 Cl and 37 Cl. Its relative atomic mass is 35.45. Calculate the percentage abundance of each isotope.

6C Hard

An element has three isotopes with mass numbers 24, 25 and 26. The abundance of the mass-25 isotope is twice that of the mass-26 isotope. The relative atomic mass is 24.32. Determine the abundance of each isotope.

2. Electronic Structure

2.1 Principal shells and subshells

7A Easy

List all the subshells that may exist in the principal shell $n = 3$.

7B Medium

Determine:
1. the number of subshells in the $n = 4$ shell,
2. the total number of orbitals,
3. the maximum number of electrons.

7C Hard

Which of the following orbital labels are impossible? 1p, 2d, 3f, 4d, 4f, 5g. For each impossible label, explain why it is not allowed.

2.2 Number and relative energies of *s*, *p* and *d* orbitals

8A Easy

State the number of orbitals and the maximum number of electrons in each of the s, p and d subshells.

8B Medium

Arrange the following orbitals in the approximate order in which they are filled in neutral atoms: 3p, 3d, 4s, 4p, 5s. Explain why orbital energy cannot always be determined using principal quantum number alone.

8C Hard — Multi-statement MCQ

Consider the following statements.
1. In a hydrogen atom, orbitals with the same principal quantum number have the same energy.
2. In a many-electron atom, a $3s$ electron generally experiences greater penetration than a $3p$ electron.
3. A $4s$ orbital is always lower in energy than a $3d$ orbital, regardless of whether the orbitals are occupied.

Which statements are correct?
A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1 and 3 only E. 1, 2 and 3

2.3 Shapes and orientations of orbitals

9A Easy

Describe the general shapes of an s orbital and a p orbital.

9B Medium

Explain why a p subshell contains three orbitals even though all three orbitals have the same general shape and energy.

9C Hard

Evaluate the following statements:
1. "A p orbital is the path followed by an electron around the nucleus."
2. "The three p orbitals differ mainly in their orientation."
3. "The nucleus lies at a node of a p orbital."

2.4 Aufbau principle

10A Easy

Use the Aufbau principle to write the ground-state electronic configuration of calcium.

10B Medium

A student proposes the configuration $1s^2 2s^2 2p^6 3s^2 3p^5 4s^1$ for the ground state of an 18-electron atom. Explain why this configuration does not represent the ground state and give the correct configuration.

10C Hard

The Aufbau principle is sometimes stated as "electrons always enter the lowest-energy orbital available." Explain why this statement is useful but incomplete. Refer to electron–electron repulsion, the closeness in energy of $4s$ and $3d$, and anomalous configurations such as chromium.

2.5 Pauli exclusion principle

11A Easy

State the Pauli exclusion principle and explain why one orbital can contain no more than two electrons.

11B Medium

Identify the error in each situation:
1. one orbital contains three electrons,
2. one orbital contains two electrons with parallel spins,
3. two electrons in the same atom are assigned an identical set of four quantum numbers.

11C Hard

Classify each configuration as violating the Aufbau principle, the Pauli exclusion principle, Hund's rule, or none of these.
1. Two electrons of identical spin occupy the same orbital.
2. Two $2p$ electrons are paired in one orbital while the other $2p$ orbitals are empty.
3. An electron occupies $4s$ while the $3p$ subshell is incomplete.
4. Five $3d$ electrons occupy five separate orbitals with parallel spins.

2.6 Hund's rule

12A Easy

Draw the orbital-box representation of a $2p^2$ configuration in its ground state.

12B Medium

Determine the number of unpaired electrons in carbon, nitrogen, oxygen and fluorine. Explain how Hund's rule is used.

12C Hard

Two possible arrangements for a p^3 configuration are proposed:
- Arrangement A: one electron in each p orbital with parallel spins.
- Arrangement B: two electrons paired in one orbital and one electron in a second orbital.

Explain why Arrangement A is lower in energy. Your explanation should go beyond simply quoting Hund's rule.

2.7 Electronic configurations of atoms and ions

13A Easy

Write the electronic configurations of:
1. S
2. S^{2-}
3. Na^{+}

13B Medium

Write the electronic configurations of:
1. Fe,
2. Fe^{2+}
3. Fe^{3+}
4. Cu^{2+}

13C Hard

A species contains 23 protons and 18 electrons.
1. Identify the species.
2. Write its full and abbreviated electronic configurations.
3. State the number of unpaired electrons.
4. Explain whether it is isoelectronic with argon.

2.8 Removal of 4*s* electrons before 3*d* electrons

14A Easy

Write the electronic configurations of Fe and Fe^{2+} . Indicate which electrons are removed during ion formation.

14B Medium

Write the electronic configurations of:
1. Mn
2. Mn^{2+}
3. Mn^{3+}

State the order in which the electrons are removed.

14C Hard

A student argues:
"Because 4*s* fills before 3*d*, 3*d* electrons must always be removed before 4*s* electrons." Explain the flaw in this argument. Discuss how orbital energies change once the 3*d* subshell becomes occupied.

2.9 Stability of half-filled and fully filled subshells

15A Easy

Write the observed ground-state configurations of chromium and copper. State how each differs from the simplest Aufbau prediction.

15B Medium

Compare the two possible configurations for chromium:
$[Ar]3d^4 4s^2$ and $[Ar]3d^5 4s^1$. Explain why the second configuration is favoured.

15C Hard

An unfamiliar atom is predicted by the simplest Aufbau treatment to have the configuration

$$[\text{core}]\ (n-1)d^9 ns^2$$

Explain what alternative configuration might occur. Discuss the balance between promotion energy, reduced pairing repulsion and stabilization of a completely filled d subshell.

3. Ionisation Energy

3.1 Definition of first and successive ionisation energies

16A Easy

Define the first ionisation energy of an element. Your definition must include the appropriate physical state, amount of substance and type of species formed.

16B Medium

Write equations, including state symbols, for the first, second and third ionisation of magnesium.

16C Hard

The first and second ionisation energies of magnesium are

$$\text{IE}_1 = 738 \text{ kJ/mol and IE}_2 = 1451 \text{ kJ/mol.}$$

Calculate the energy required to convert 0.250 mol of gaseous magnesium atoms completely into gaseous Mg^{2+} ions.

3.2 Nuclear charge

17A Easy

Explain why the first ionisation energy of magnesium is greater than that of sodium.

17B Medium

Arrange the following isoelectronic species in increasing ionisation energy:

$$O^{2-}, F^-, Na^+, Mg^{2+}.$$

Explain your reasoning in terms of nuclear charge.

17C Hard

Compare the energy required for the processes

$$B^+(g) \longrightarrow B^{2+}(g) + e^-$$

and

$$C^+(g) \longrightarrow C^{2+}(g) + e^-.$$

Account for both nuclear charge and the electronic configurations of the ions involved.

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3.3 Shielding

18A Easy

Explain why first ionisation energy generally decreases from lithium to sodium to potassium.

18B Medium

A $2s$ electron and a $2p$ electron are in the same principal shell. Explain why the $2s$ electron generally experiences greater penetration, less effective shielding and stronger attraction to the nucleus.

18C Hard — **Multi-statement MCQ**

Consider the following statements.
1. Inner-shell electrons generally shield outer-shell electrons more effectively than electrons in the same shell.
1. An s electron generally penetrates closer to the nucleus than a p electron with the same principal quantum number.
1. Each inner electron reduces the nuclear attraction experienced by an outer electron by exactly one proton charge. Which statements are correct?
A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1 and 3 only E. 1, 2 and 3

3.4 Distance from the nucleus

19A Easy

Explain why the first ionisation energy of potassium is lower than that of sodium.

19B Medium

Suppose a $3s$ electron and a $4s$ electron experience approximately the same effective nuclear charge. Which electron would be easier to remove? Explain using average distance from the nucleus.

19C Hard

A hydrogen atom is promoted from its ground state to an excited state in which its electron occupies a shell with a larger principal quantum number.

Explain how this affects:

1. the average electron–nucleus distance,
2. the attraction between the electron and nucleus,
3. the ionisation energy.

Answer

In the excited state, the average electron–nucleus distance increases and the electrostatic attraction becomes weaker. The electron also has a higher energy and lies closer to the ionisation limit. Consequently, less additional energy is required to remove it, so the ionisation energy decreases.

3.5 Electron–electron repulsion

20A Easy

Explain why removing an electron from a negatively charged ion is generally easier than removing one from the corresponding neutral atom.

20B Medium

Compare the attraction experienced by the electron in He^+ with that experienced by either electron in neutral helium. Explain the role of electron–electron repulsion.

20C Hard

The second ionisation energy of an element is always greater than its first ionisation energy. Explain why, referring to ionic charge, electron–electron repulsion and attraction between the remaining electrons and the nucleus.

3.6 Subshell effects

21A Easy

Explain why the first ionisation energy of aluminium is lower than that of magnesium despite aluminium having a greater nuclear charge.

21B Medium

Explain why the first ionisation energy of boron is lower than that of beryllium. Compare the energies and penetration of the $2s$ and $2p$ subshells.

21C Hard

The first ionisation energy of gallium is not dramatically lower than that of aluminium, even though gallium lies below aluminium in Group 13. Suggest an explanation involving the relatively poor shielding provided by the filled $3d$ subshell.

3.7 Paired-electron repulsion

22A Easy

Explain why the first ionisation energy of oxygen is lower than that of nitrogen.

22B Medium

Predict and explain the corresponding anomaly between phosphorus and sulfur.

22C Hard

Consider four hypothetical atoms whose outer configurations are

$$np^1, np^2, np^3, np^4.$$

Assume that the electrons experience approximately the same effective nuclear charge and belong to the same principal shell. Predict which configuration would show an unexpectedly low ionisation energy relative to the preceding configuration. Explain using orbital occupancy and pairing repulsion.

3.8 Identifying groups and periods from successive ionisation energies

23A Easy

An element in Period 3 has the following successive ionisation energies: 738, 1451, 7733, 10540 kJ/mol.

1. Identify the major discontinuity.
2. Determine the group of the element.
3. Identify the element.

23B Medium

A successive-ionisation-energy plot shows major jumps after removal of the first and ninth electrons.

Determine:

1. the total number of electrons in the atom,
2. its electronic configuration,
3. its group and period,
4. its likely identity.

23C Hard

An unknown main-group atom shows:

- a major jump after the seventh electron is removed,
- another major jump after the fifteenth electron is removed.

Deduce the number of valence electrons, the total number of electrons, the full electronic configuration, the group and period, and the identity of the element.

3.9 Explaining apparent anomalies in periodic trends

24A Easy

Across Period 3, first ionisation energy generally increases. Explain why the first ionisation energy of aluminium is nevertheless lower than that of magnesium.

24B Medium

Explain why nitrogen has a higher first ionisation energy than oxygen even though oxygen has the greater nuclear charge.

24C Hard

Across Period 2, first ionisation energy generally increases. Identify and explain the two principal exceptions. For each exception, state whether subshell energy or paired-electron repulsion is the dominant explanation, and explain why the general increasing trend still remains valid.

4. Olympiad Extension: Mass Spectrometry

The following questions provide the necessary principles in their stems. Students are assessed on interpretation, deduction and calculation rather than prior knowledge of instrumental design.

4.1 Reading an isotope spectrum

MS1 Reading an isotope spectrum

A mass spectrum separates singly charged ions according to their mass-to-charge ratio, $\frac{m}{z}$. For singly charged ions, $\frac{m}{z}$ is numerically equal to the isotope's relative mass.

The spectrum of element X contains the following peaks:

$\frac{m}{z}$	Relative intensity
24	79
25	10
26	11

1. State the number of naturally occurring isotopes shown.
2. Calculate the relative atomic mass of X.
3. Suggest the identity of X.

4.2 Time-of-flight reasoning

MS2 Time-of-flight reasoning

In a time-of-flight mass spectrometer, ions of charge q are accelerated from rest through the same potential difference V. The electrical energy gained becomes kinetic energy:

$$qV = \tfrac{1}{2}mv^2.$$

The time required to travel through a field-free tube of length L is

$$t = \frac{L}{v}.$$

1. Derive an expression for t in terms of m, q, V and L.
2. Two singly charged isotopes have masses 25 and 100. Determine the ratio of their flight times.
3. Explain why heavier ions arrive later.

4.3 The meaning of mass-to-charge ratio

MS3 The meaning of m/z

A mass spectrometer does not measure mass directly. It measures the mass-to-charge ratio m/z , where z is the number of positive elementary charges on the ion. Two peaks are produced by $^{20}Ne^{+}$ and $^{40}Ar^{2+}$.

1. Calculate the m/z value of each ion.
2. Explain why the two ions appear at the same position.
3. Explain why a single peak at $m/z = 20$ is not sufficient evidence that an ion has a mass of 20.

4.4 Chlorine isotope patterns

MS4 Chlorine isotope patterns

Chlorine has two common isotopes:

- ^{35}Cl, abundance 75%
- ^{37}Cl, abundance 25%

A Cl_2 molecule can therefore contain two ^{35}Cl atoms, one of each isotope, or two ^{37}Cl atoms.

Determine:

1. the masses of the three possible Cl_2^{+} molecular ions
2. their expected relative probabilities,
3. the approximate ratio of their peak intensities.

4.5 Distinguishing isotope abundance from instrumental response

MS5 Evaluating a conclusion

A mass spectrum shows two peaks at $m/z = 79$ and $m/z = 81$ with nearly equal intensities.

A student concludes:

"The two isotopes must have exactly equal natural abundances." Evaluate this conclusion.

4.6 Instrumental resolution

MS6 Instrumental resolution

The resolving power R of a mass spectrometer is defined as

$$R = \frac{m}{\Delta m},$$

where m is the approximate mass and Δm is the smallest mass difference that can be distinguished. Two ions have masses 28.00 and 28.03.

1. Calculate the minimum resolving power needed to separate them.
2. State whether an instrument with $R = 500$ can distinguish the peaks.
3. State whether an instrument with $R = 1200$ can distinguish the peaks.

Answer

$$\Delta m = 28.03 - 28.00 = 0.03.$$

Using $m \approx 28.00$,

$$R = \frac{28.00}{0.03} = 933.3.$$

The minimum whole-number resolving power is therefore approximately

$$\boxed{R = 934}.$$

An instrument with $R = 500$ cannot resolve the peaks, whereas an instrument with $R = 1200$ can.

5. Olympiad Extension: Photoelectron Spectroscopy

In photoelectron spectroscopy (PES), photons eject electrons from gaseous atoms. The energy relationship is

$$E_{\text{photon}} = E_{\text{binding}} + E_{\text{kinetic}}.$$

A simplified PES spectrum is interpreted as follows:

- peak position indicates electron binding energy;
- peak area is proportional to the number of electrons in the corresponding subshell;
- a higher binding energy means that an electron is held more strongly.

The questions below include all information needed for reasoning.

5.1 Identifying subshells

PES1 Identifying subshells

An atom produces three PES peaks:

Peak	Relative binding energy	Relative area
A	very high	2
B	medium	2
C	low	1

1. Assign each peak to a subshell.
2. Write the electronic configuration.
3. Identify the element.

5.2 Using peak areas

PES2 Using peak areas

An unknown atom has five PES peaks with relative areas

$$2 : 2 : 6 : 2 : 3.$$

The peaks are listed from highest to lowest binding energy.

1. Assign each peak to a subshell.
2. Determine the total number of electrons.
3. Identify the atom.
4. State the number of unpaired electrons in its ground state.

5.3 The magnesium–aluminium anomaly

PES3 Explaining the magnesium–aluminium anomaly

Simplified PES data for the valence electrons of magnesium and aluminium are shown below.

Element	Valence subshell	Relative binding energy
Mg	$3s$	higher
Al	$3p$	lower

Both elements contain inner electrons with similar shielding.

1. Which element should have the lower first ionisation energy?
2. Explain why.
3. Explain why this does not contradict aluminium having the greater nuclear charge.

5.4 Comparing the same subshell across a period

PES4 Binding-energy shifts across a period

Two elements X and Y have valence configurations

$$X : 3s^1,\ Y : 3s^2.$$

The 3s PES peak of Y occurs at a higher binding energy than that of X.
1. Which element has the greater nuclear charge?
2. Identify X and Y if they are consecutive Period 3 elements.
3. Explain the shift in binding energy.

5.5 Paired-electron repulsion

PES5 Paired-electron repulsion

Nitrogen and oxygen have the valence configurations

$$\mathrm{N} : 2\mathrm{p}^3,\ \mathrm{O} : 2\mathrm{p}^4$$

. In oxygen, one 2p orbital contains two electrons.
1. Why is one of oxygen's 2p electrons easier to remove than might be expected from the increase in nuclear charge?
1. How does this explain the first-ionisation-energy anomaly between nitrogen and oxygen?
2. Would the total area of oxygen's 2p peak be greater or smaller than nitrogen's?

5.6 Identifying an ion from its spectrum

PES6 Identifying an ion

A gaseous species produces PES peaks corresponding to $1s^2 2s^2 2^6$. The nucleus of the species contains 12 protons.
1. Determine the number of electrons.
2. Determine the charge of the species.
3. Identify the species.
4. Predict whether its electrons would have higher or lower binding energies than those of neutral neon.

5.7 Distinguishing ground and excited states

PES7 Ground and excited states

An atom normally has the ground-state configuration $1s^2 2s^2 2p^6 3s^2$. After absorbing energy, one electron is promoted, producing $1s^2 2s^2 2p^6 3s^1 3p^1$. A simplified PES experiment is carried out before the atom relaxes.
1. How would the number of occupied valence subshells change?
2. Which electron would have the lower binding energy: the 3s or 3p electron?
3. Would the excited atom be easier or harder to ionise than the ground-state atom?

5.8 Comparing mass spectrometry and PES

PES8 Comparing what MS and PES reveal

Two experimental results are obtained for an unknown element.
Mass spectrum: the element has isotopes with masses 34, 36 and 38.
PES spectrum: the peak areas correspond to
$1s^2 2s^2 2p^6 3s^2 3p^4$.
1. What information does the mass spectrum provide?
2. What information does the PES spectrum provide?
3. Identify the element.
4. Explain why the mass spectrum alone cannot determine the ground-state electronic configuration.

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